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RogueGPT: A Controlled Stimulus Generation Framework for News Authenticity Research

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Software

- Review
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Summary

RogueGPT is an open-source Python framework for the controlled, reproducible generation and curation of multilingual news fragments for AI authenticity research. It enables researchers to systematically produce synthetic news stimuli across a wide range of large language model (LLM) families, journalistic styles, languages, and content formats, while storing every fragment alongside complete generation provenance in a MongoDB corpus.

The framework follows a three-layer architecture that strictly separates the data logic (`core.py`) from user-facing interfaces: a Streamlit web application, a command-line interface (CLI), and a Model Context Protocol (MCP) server for AI-agent integration. All three interfaces share a single validation and normalisation layer, ensuring that every fragment — whether ingested manually, via automated generation scripts, or through an AI agent — conforms to the same schema and that all experimental parameters are recorded for reproducibility.

The current corpus contains more than 2,600 multilingual news fragments generated by 37 model configurations across 10 providers (OpenAI, Anthropic, Google, Meta, Mistral, DeepSeek, Microsoft, Zhipu, Moonshot, and Qwen), covering four languages (English, German, French, and Spanish), three content formats (tweet, headline, and short article), and five journalistic styles per language. Alongside the machine-generated material, 222 human-authored fragments — both legitimate and fake — serve as experimental anchors for perception studies.

RogueGPT is the upstream stimulus generation component of a three-tool research pipeline: CRED-1 (Loth et al., 2025) identifies unreliable news sources, RogueGPT generates controlled stimuli, and JudgeGPT delivers them to human participants for perception measurement.

Statement of Need

Empirical research on human and automated detection of AI-generated news requires stimuli that are diverse, systematically varied across experimental conditions, and fully reproducible. Existing corpora are typically static snapshots generated by a single model or a narrow set of models, lacking the fine-grained metadata required to isolate the effect of individual experimental variables (model family, version, prompt strategy, language, style) on detection outcomes (Kreps et al., 2022; Zellers et al., 2019).

RogueGPT addresses this gap in three ways:

1. **Systematic parameterisation.** Stimuli are generated by Cartesian expansion of a declarative configuration (`prompt_engine.json`) that defines models, languages, styles, and formats. Every combination is reachable without manual intervention, enabling researchers to expand the corpus to new model releases in minutes by updating a single JSON file.

2. **Complete provenance.** Every fragment is stored with the exact model identifier, prompt text, ISO language code, content format, journalistic style, ingestion channel, and a UTC creation timestamp. This makes it possible to reproduce any stimulus exactly and to filter the corpus along any experimental axis at query time via the CLI or MCP API.
3. **Living corpus design.** Unlike frozen datasets, the corpus is designed to grow. The MCP server exposes `ingest_fragment` and `retrieve_fragments` as tool-use endpoints, enabling AI agents to autonomously extend the corpus with new model outputs as part of larger automated pipelines. This is particularly important given the rapid pace of LLM releases: a framework that supports incremental corpus expansion without re-running entire generation pipelines is essential for longitudinal research.

The framework is part of a broader open-source research ecosystem for studying online misinformation. Upstream, the CRED-1 Domain Credibility Dataset (Loth et al., 2025) provides the automated source selection that identifies unreliable domains whose content is scraped as human-authored fake news anchors. Downstream, JudgeGPT delivers RogueGPT-generated stimuli to human participants and records dual-axis perception judgements. Together, the three tools form an end-to-end pipeline from source identification through stimulus generation to quantitative perception measurement.

RogueGPT is already in active use in peer-reviewed publications from the same research programme:

- *Industrialized Deception* (Loth, Kappes, et al., 2026d) analyses the systemic effects of LLM-generated misinformation on digital ecosystems, using stimuli generated by RogueGPT (ACM WWW '26).
- *Eroding the Truth-Default* (Loth, Kappes, et al., 2026a) reports human perception data from 320 participants who evaluated stimuli produced by this framework (ACM WWW '26).
- *The Verification Crisis* (Loth, Kappes, et al., 2026b) presents expert survey findings on GenAI disinformation and reproducible provenance as part of the same research programme (ACM WWW '26).
- The foundational survey *Blessing or Curse?* (Loth et al., 2024) describes the methodology underpinning the generation pipeline.

The full corpus produced by RogueGPT is archived on Zenodo (DOI: 10.5281/zenodo.18703138) and available for academic research under restricted access.

Key Features

Multi-Model Support

The model registry in `prompt_engine.json` currently defines 37 identifiers across 10 providers. The naming convention (`provider_model-variant`) is provider-agnostic; adding a new model requires a single line change to the JSON file and no code modifications. At ingest time, the CLI and MCP server validate submitted model identifiers against the registry, with an optional `--lenient` flag for provisional use of unreleased models.

Structured Fragment Schema

Each fragment is stored with complete provenance metadata including model identifier, prompt text, ISO language code, content format, journalistic style, source outlet, and UTC timestamp. A shared validation layer enforces schema consistency across all three interfaces, ensuring that every fragment — regardless of ingestion path — carries the metadata required to reproduce experimental conditions exactly. The full schema specification is documented in the repository.

Interfaces

RogueGPT provides three user-facing interfaces that share a single validation layer: a command-line interface for scripted ingestion and retrieval, a Streamlit web application for interactive use, and a Model Context Protocol (MCP) server that enables LLM-based agents to query the corpus and ingest new fragments programmatically. The MCP integration allows AI agents to autonomously extend the corpus as part of larger automated research pipelines.

Human Fragment Sourcing

Human-written fragments are sourced from two pools:

- **Legitimate news:** Opening paragraphs from established international outlets (BBC, Guardian, Reuters, Tagesschau, FAZ, Le Monde, El País) via RSS feeds.
- **Fake news:** Articles scraped from domains classified as *fake*, *unreliable*, or *conspiracy* by the CRED-1 Domain Credibility Dataset (Loth et al., 2025), which scores 2,672 domains using editorial classification, fact-check claims, web popularity, and domain age.

Both categories are ingested with full provenance metadata, providing experimental anchors that reflect authentic disinformation language rather than artificially constructed examples.

State of the Field

Several tools and datasets address fake news generation and detection research, but none provide an active, extensible generation infrastructure with full provenance tracking.

GROVER (Zellers et al., 2019) generates and detects neural fake news but is limited to a single GPT-2-scale architecture and English text. The LIAR dataset (W. Y. Wang, 2017) provides 12,836 human-labelled claims across six fine-grained labels but offers no generation infrastructure. TruthfulQA (Lin et al., 2022) measures model truthfulness across 817 questions but targets factual consistency rather than controlled stimulus generation. Kreps et al. (Kreps et al., 2022) demonstrate that GPT-2-generated political news is perceived as credible as human-written content, establishing the need for systematic multi-model comparisons — but their work uses a single model without a reusable generation framework.

More recent efforts such as the M4 corpus (Y. Wang et al., 2024) collect multi-generator outputs across domains and languages but distribute them as static datasets without tooling for reproducible expansion. Similarly, MULTITuDE (Macko et al., 2023) covers 11 generators across 11 languages but is frozen at its release configuration. FakeNewsNet (Shu et al., 2020) provides a rich repository combining news content with social context but focuses on detection research with pre-collected articles rather than controlled stimulus generation.

RogueGPT differs from all of these in three respects: (1) it is an active generation infrastructure rather than a frozen benchmark, supporting incremental corpus expansion as new models are released; (2) it stores complete generation provenance per fragment, enabling fine-grained experimental filtering by model, language, style, and format; and (3) its MCP integration is, to our knowledge, the first such interface for a research corpus management system, enabling direct use within AI agent workflows.

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